

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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Build cost-optimized storage with Amazon S3

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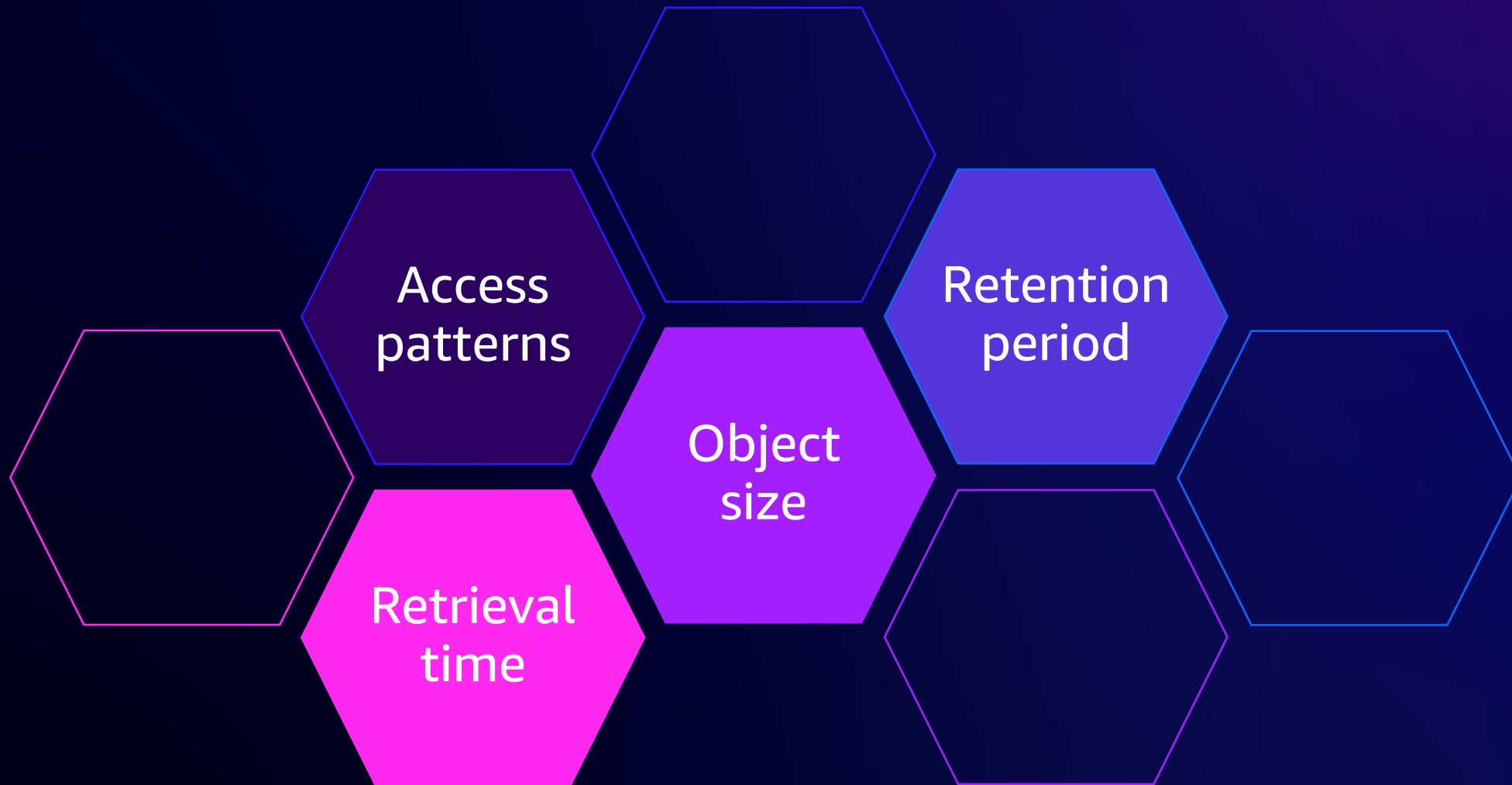
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AWS



Agenda

- 01 Identify your workload requirements
- 02 Pick the right storage class for your workload
- 03 Optimize your costs with Amazon S3 tools
- 04 Develop insights into your Amazon S3 usage
- 05 Discussion

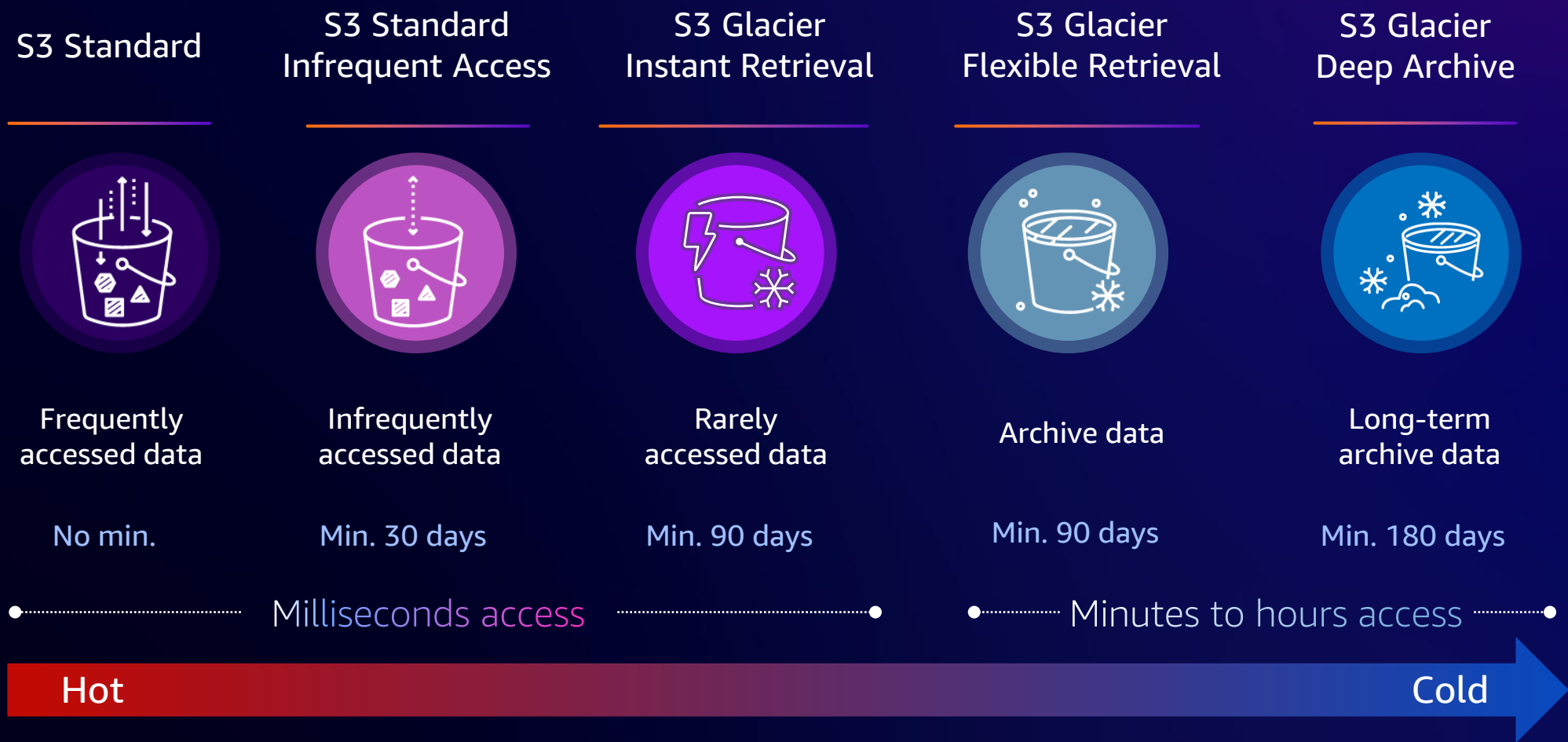
Workload-based cost optimization



Choose the right Amazon S3 storage class for your workload



Storage class choice matters at scale



Examples

- Storage with no activity for first 90 days (Storage Only, 100% COLD)

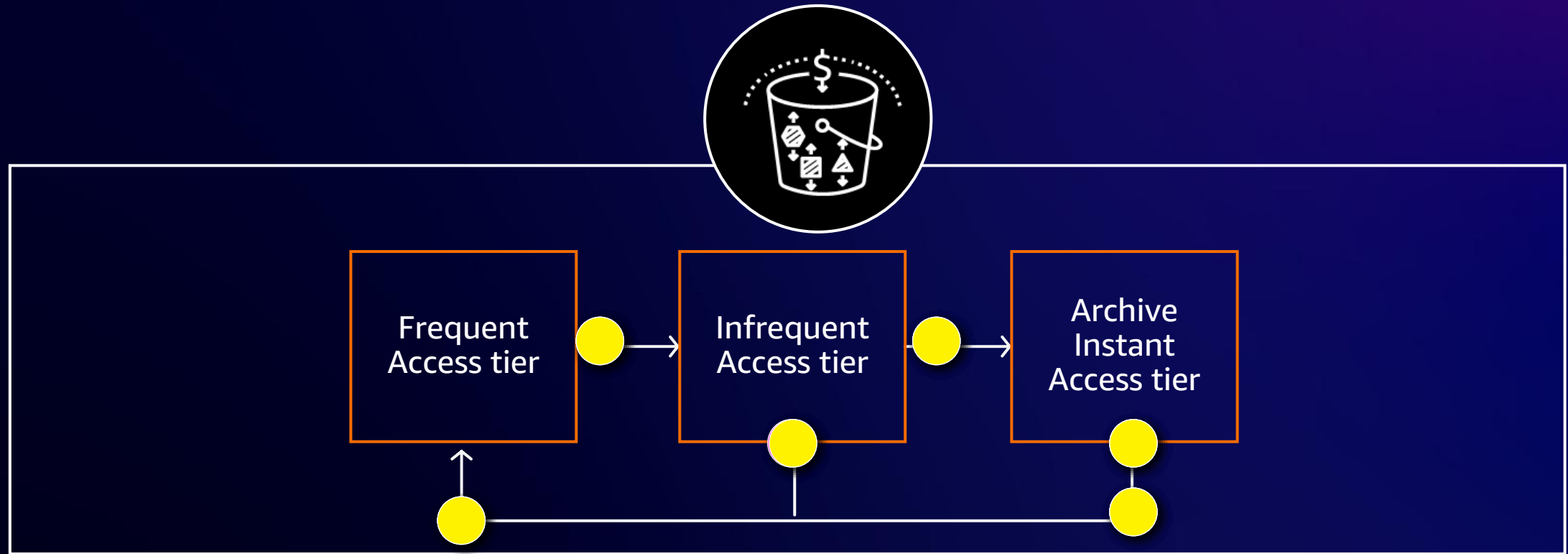
Storage \$ Comparison	S3 Standard	S3 Infrequent Access	S3 Glacier Instant Retr	S3 Glacier Flex retr	S3 Glacier Deep Arch
\$/GB	0.023	0.0125	0.004	0.00036	0.00099
% to S3 Std.		-46%	-83%	-84%	-96%

- Tier 1 request costs

PUT Request \$ Comparison	S3 Standard	S3 Infrequent Access	S3 Glacier Instant Retr	S3 Glacier Flex retr	S3 Glacier Deep Arch
\$/1000 Reqs	0.005	0.01	0.02	0.03	0.05
% to S3 Std		+100%	+300%	+500%	+900%

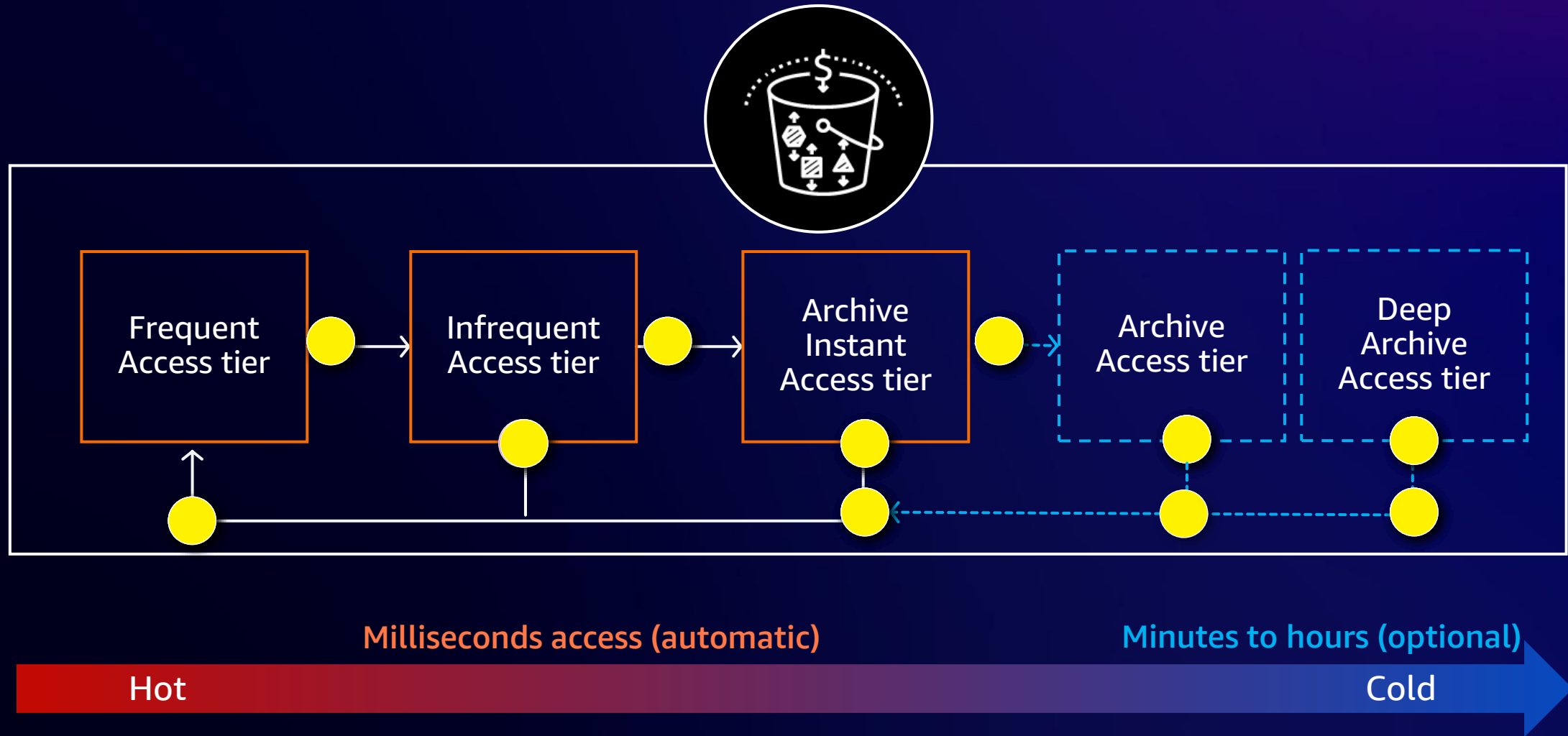
*All numbers are for US-east(N. Virginia) region.

Automatic cost optimization with S3 Intelligent-Tiering



Milliseconds access (automatic)

Automatic cost optimization with S3 Intelligent-Tiering



\$2 billion
in savings



Optimize costs with Amazon S3 tools

Amazon S3 Lifecycle helps optimize storage costs



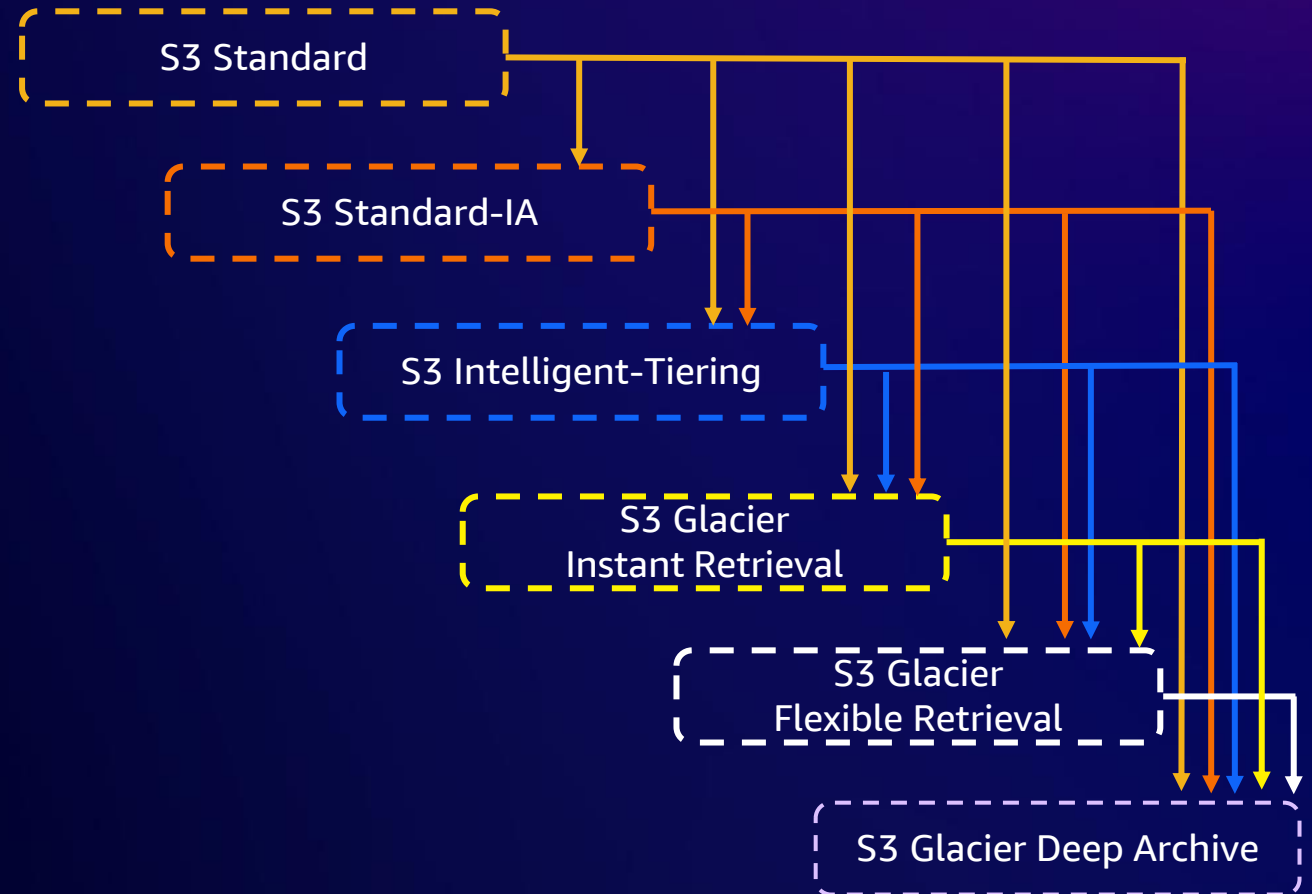
Transition actions



Expiration actions



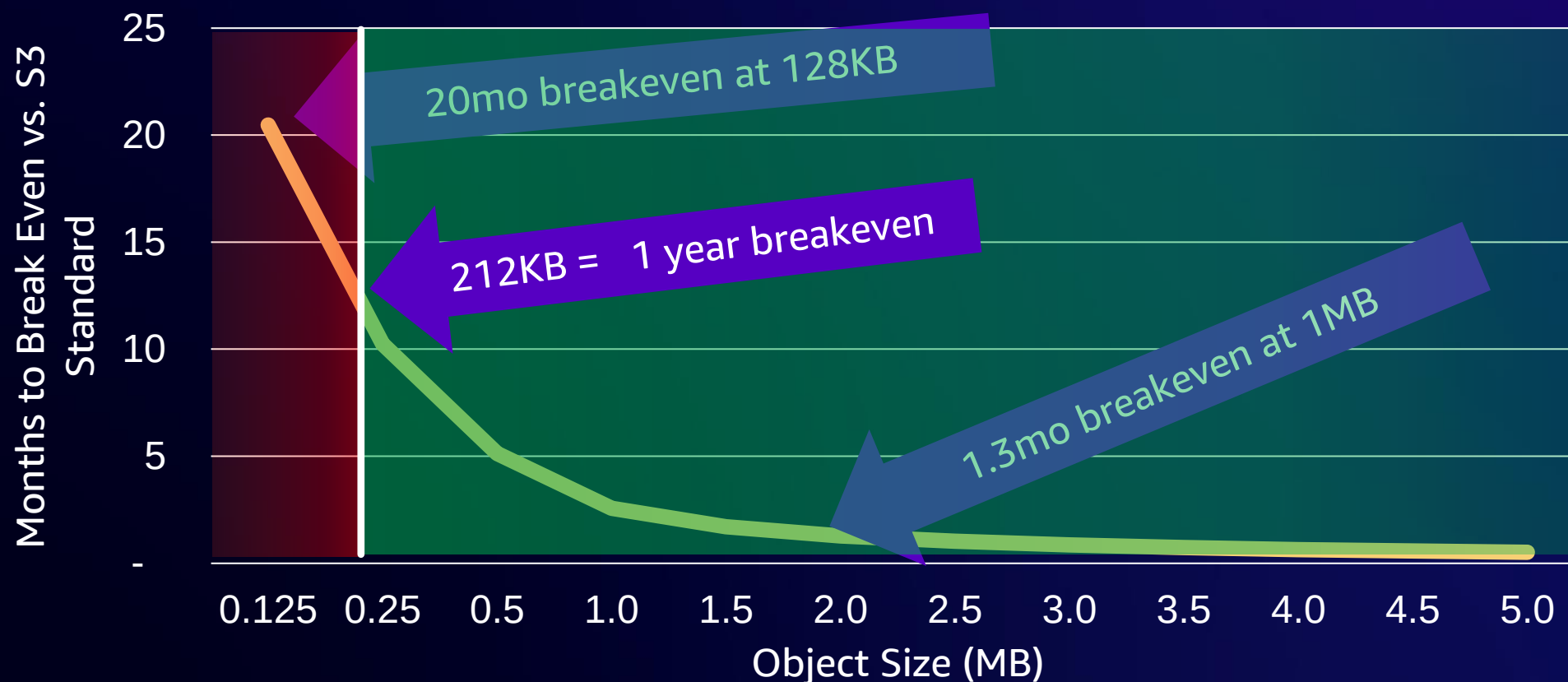
Action filters



Retention time vs. one-time lifecycle costs

Breakeven = "Lifecycle API Costs" / ("S3 Std \$ per mo" – "GDA \$ per mo")

S3 Glacier Deep Archive Break Even Months vs. S3 Standard



*All numbers are for US-east(N. Virginia) region.

If you can't see it,
you can't change it

Analyze your storage with S3 Storage Lens



Free and paid versions available

1. Interactive dashboard experience in the Amazon S3 console – **free to all** customers
2. Organization-wide visibility
3. Drill down by Region, storage class, bucket, and prefix
4. Multiple ways to access S3 Storage Lens data
5. Granular usage and activity metrics

Find the answers with S3 Storage Lens



28 free metrics
for instant visibility

- Are we using any of the lower-cost Amazon S3 storage classes?
- Do we have any incomplete multipart uploads > 7 days?
- How much of my storage is comprised of non-current versions of objects?



35 advanced metrics
to optimize cost

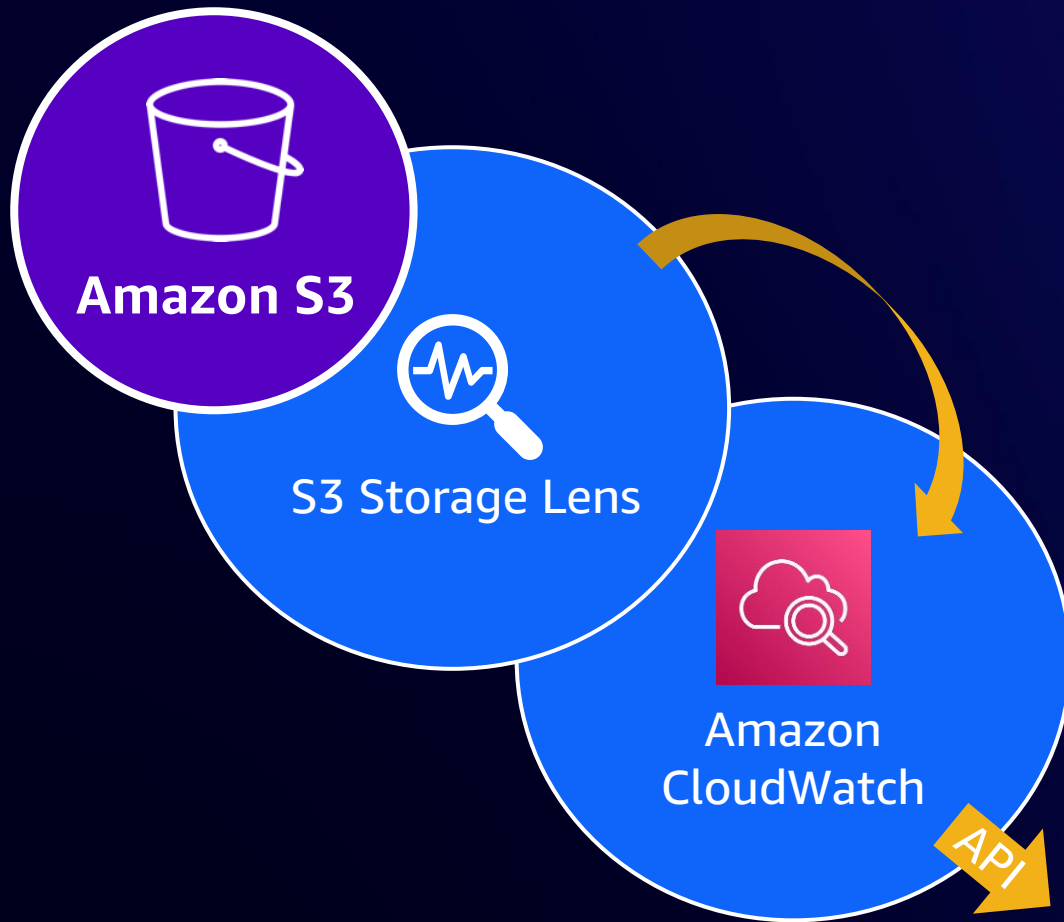
- Which of my buckets are not accessed frequently?
- Which buckets have dynamic access patterns?
- Which buckets don't have a lifecycle policy?

Tutorial:

Getting started with **Amazon S3 Storage Lens**



Publish metrics to Amazon CloudWatch



Unlock capabilities of CloudWatch for S3 Storage Lens metrics

- Centralized, cross-service dashboards
- Alarms and triggered actions
- Anomaly detection

Access metrics via CloudWatch API

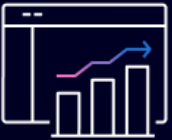
- Connect AWS observability partners



Monitor S3 costs and set up alarms



Configure **10 free** Amazon S3 cost and usage reports per account in the AWS Billing console



Monitor your Amazon S3 cost reports **over time**



Set up billing **alarms and notifications** using Amazon CloudWatch

Key takeaways



Use the right storage class based on your workload requirements



Gain insights with S3 Storage Lens and AWS Billing reports



Continuously monitor your Amazon S3 usage and costs

Continue your AWS Storage learning

Take a
learning plan



Set your AWS Storage
Learning Plans via
AWS Skill Builder

Increase your
knowledge



Use our **Ramp-Up Guides**
to build your storage
knowledge

Earn AWS
Storage badges



Demonstrate your
knowledge by achieving
digital badges

aws.training/storage

*Scan to go directly to the Storage
Knowledge Badge Readiness Paths:*



Thank you!

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Please complete the session
survey in the mobile app

